Majorana Modes in the Machine

Week 5: Transitioning to Quantum Circuit Measurement

Dobromir Stoev
Edgar Harutyunyan
Guilherme Schewtschik
Jaskaran Singh
Zhengyi Liu
Zhenming Shi

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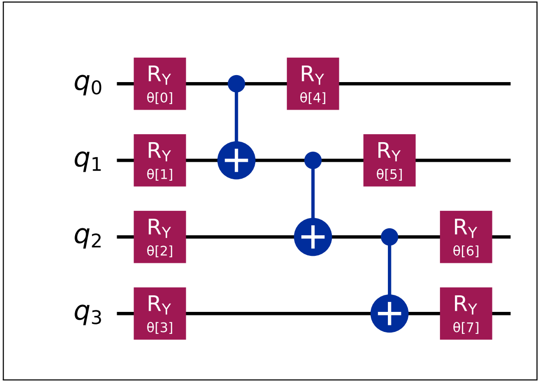
Week 5 Objectives: Moving to Block 3

- **Main Goal:** Shift from "Exact Matrix Math" to "Gate-Based Simulation."
- **Task 1:** Ground State Preparation via VQE (Variational Quantum Eigensolver).
- **Task 2:** Defining the Non-local String Order Parameter $\mathcal{O}_{string}$.
- **Task 3:** Mapping Majorana Observables to Qubit Measurement Gates.
- **Task 4:** Numerical Evidence.

Task 1: VQE Circuit Ansatz (Visual)

Hardware-Efficient Ansatz:

- Uses only R_y to keep the state in the real domain.
- Linear CNOT entanglement mimics 1D hopping.



R_y Gate

For $L = 4$, the configuration space consists of $2^4 = 16$ possible states:

$$|\Psi\rangle = c_0|0000\rangle + c_1|0001\rangle + \cdots + c_{15}|1111\rangle$$

Initially, $|\psi\rangle = [1, 0, 0, \ldots, 0]^T$. An $R_y(\theta)$ gate on q_0 performs:

$$R_y(\theta)|0\rangle = \cos(\theta/2)|0\rangle + \sin(\theta/2)|1\rangle$$

In the 16D vector, the value "1" splits:

- Index 0 ($|0000\rangle$): becomes $\cos(\theta/2)$
- Index 8 ($|1000\rangle$): becomes $\sin(\theta/2)$

CNOT Gate

Entanglement occurs when a gate moves amplitudes between binary configurations.

CNOT(0,1) Operation

If q_0 is 1, flip q_1. In the vector:

1. $|1000\rangle$ (Index 8) satisfies the condition.
2. It is transformed into $|1100\rangle$ (Index 12).

Physical meaning: isolated excitations are combined into **paired states**, building the topological correlations needed for Majoranas.

Hilbert Space Evolution: Mathematical Verification

Mathematical Constraint: Post-entanglement states cannot be factored into independent single-qubit probabilities. The evolution must be tracked across the full $2^4 = 16$ dimensional state vector.

State Vector Evolution Table (Tracking non-zero amplitudes):

Index (Dec)	**Basis ($q_0 q_1 q_2 q_3$)**	**Init State** $\lvert\psi_0\rangle$	**Post** $R_y(\theta_0)$ $\lvert\psi_1\rangle$	**Post CNOT$_{0\to1}$** $\lvert\psi_2\rangle$
0	$\lvert 0000\rangle$	1	$\cos(\frac{\theta_0}{2})$	$\cos(\frac{\theta_0}{2})$
1	$\lvert 0001\rangle$	0	0	0
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
8	$\lvert 1000\rangle$	0	$\sin(\frac{\theta_0}{2})$	**0** *(Amplitude shifted)*
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
12	$\lvert 1100\rangle$	0	0	$\sin(\frac{\theta_0}{2})$

Hamiltonian Simplification: The Pure YY Limit

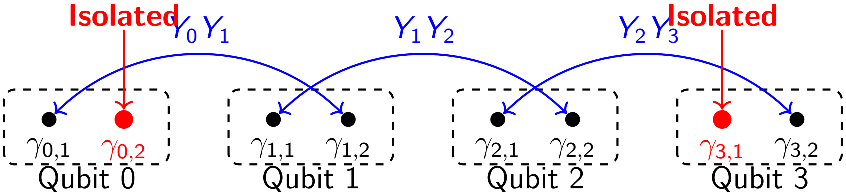
General Qubit Mapping (Block 2):

$$H_K = -\frac{\mu}{2}\sum Z_j + \frac{t-\Delta}{2}\sum X_j X_{j+1} + \frac{t+\Delta}{2}\sum Y_j Y_{j+1}$$

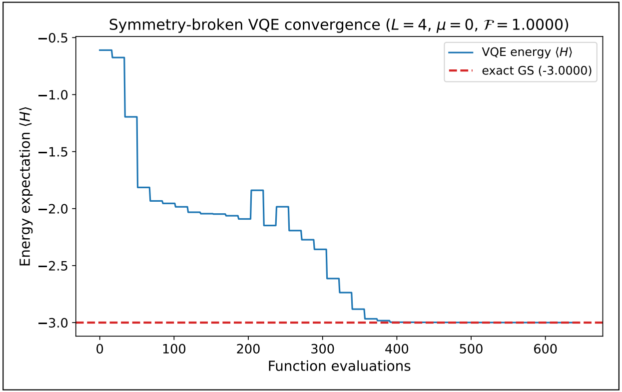
Topological Sweet Spot ($t = \Delta$, $\mu = 0$):

$$\mathbf{H} = \sum_{\mathbf{j=0}}^{\mathbf{L-2}} \mathbf{Y_j Y_{j+1}} \equiv \mathbf{i} \sum_{\mathbf{j=0}}^{\mathbf{L-2}} \gamma_{\mathbf{j,1}} \gamma_{\mathbf{j+1,2}}$$

- **Algebraic Cancellation:** Setting $t = \Delta$ strictly eliminates all $X_j X_{j+1}$ components.
- **Pairing Shift Geometry:** The pure $Y_j Y_{j+1}$ interaction fundamentally alters the topology by coupling the **left mode ($\gamma_{j,1}$)** to the **right mode ($\gamma_{j+1,2}$)**, effectively isolating the inner edge modes.



Numerical Verification: VQE Convergence Dynamics



Optimization Parameters:

- **Algorithm:** L-BFGS-B (Gradient-based).
- **Cost Function:** $C(\vec{\theta}) = \langle H \rangle - \lambda \langle P \rangle$.
- **Penalty ($\lambda = 0.1$):** Enforces strictly even parity ($P = +1$) selection, breaking topological degeneracy.

Task 2: String Operator

Boundary Majorana Correlator to Qubit Mapping:

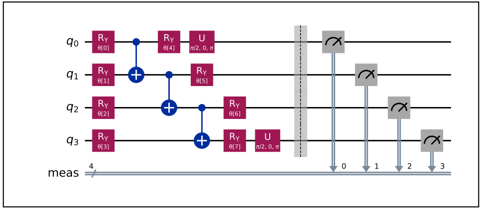
$$\hat{O}_{corr} \propto i\gamma_{0,2}\gamma_{3,1} = i(Y_0)(Z_0 Z_1 Z_2 X_3) = -X_0 Z_1 Z_2 X_3$$

- **1. Breakdown of the Landau Paradigm:** The topological phase hosts no local symmetry breaking. All local single-qubit expectations strictly vanish:

  $$\forall j \in \{0, 1, 2, 3\}, \quad \langle X_j \rangle = 0, \quad \langle Y_j \rangle = 0, \quad \langle Z_j \rangle = 0$$

  Non-local order parameters are mandatory to detect this hidden order.
- **2. Ground State Commutation Constraint:** A saturated non-zero invariant requires $[H, \hat{O}_{string}] = 0$. The naive boundary operator $\hat{O}_{naive} = X_0 X_3$ anti-commutes with H, creating high-energy quasiparticle excitations and yielding $\langle X_0 X_3 \rangle \equiv 0$.

Task 3: Basis Rotation via Hadamard Transform



Target Observable: $\hat{O}_{corr} = -X_0 Z_1 Z_2 X_3$

Algebraic Mapping ($HXH = Z$):

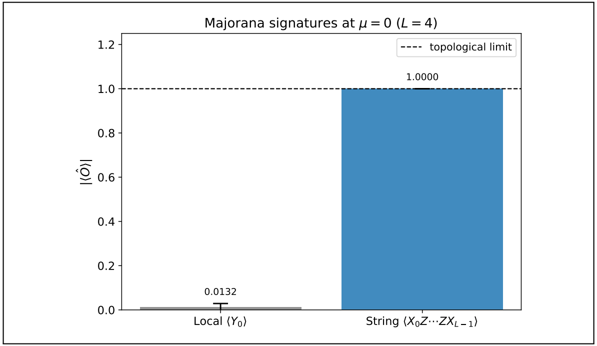
$$
\begin{aligned}
HXH &= \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}\\
&= \frac{1}{2}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}\begin{pmatrix}1 & -1\\ 1 & 1\end{pmatrix}\\
&= \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix} \equiv Z
\end{aligned}
$$

Hardware Constraint:

Superconducting qubits are strictly restricted to Z-basis projection. Majoranas require X-basis detection.

- H is applied specifically to q_0 and q_3 to reconstruct the non-local string.

Task 4: Numerical Evidence: The 1.0 Correlation



- **Left (Y_0):** ≈ 0 due to anti-commutation with the global parity operator ($\{Y_0, \hat{P}\} = 0$).
- **String ($X_0 Z_1 Z_2 X_3$):** Exactly 1.0000 (Theoretical limit).
- **Algebraic Symmetry:** The commutation $[H, \hat{O}_{corr}] = 0$ identifies this as a conserved topological invariant.
- **Result:** Non-local Topology confirmed in the gate-based model.

Summary & Next Steps

- **Completed:** VQE ground state preparation.
- **Discovery:** The non-local string order parameter is a robust topological signature.
- **Next:** Sweep μ to generate the full Phase Diagram using only quantum circuit results.